
Measurement Invariance of the DDI Dermatology Benchmark Across Fitzpatrick Skin Type

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Abstract

1 Fairness audits of dermatology AI typically compare accuracy across Fitzpatrick
2 Skin Type (FST) groups—a six-level dermatologic scale of constitutive skin
3 pigmentation, conventionally collapsed to FST-I/II (lightest), III/IV, and V/VI
4 (darkest)—and treat the gap as direct evidence of differential model capability.
5 This inference confuses three distinct phenomena: differences in capability, differ-
6 ences in item difficulty, and differences in how items function as measurements
7 across groups. We propose to disentangle these by combining amortized Item
8 Response Theory (IRT) calibration of foundation vision-language models against
9 the Diverse Dermatology Images (DDI) benchmark with item-level measurement-
10 invariance analysis grouped by FST. Our primary endpoint is an aggregate FST
11 difficulty shift on the Rasch logit scale, with a pre-specified *falsification threshold*—
12 a magnitude floor below which we will declare no meaningful non-invariance even
13 if a statistical test rejects—of $|\Delta b_{V-VI-I-II}| \geq 0.5$ logits, conjoined with a boot-
14 strap 95% CI excluding zero. A Linear Logistic Test Model (LLTM) decomposes
15 whether residual FST signal survives controls for lesion category and malignancy.

16 1 Introduction

17 Dermatology AI tools are increasingly deployed in clinical care, with a wider range of applications
18 under active consideration [Daneshjou et al., 2022]. Fitzpatrick-Skin-Type-stratified accuracy has
19 become the dominant fairness metric in those evaluations [Daneshjou et al., 2022, Groh et al., 2021],
20 and regulators, procurement reviewers, and journals now treat subgroup gaps as direct evidence of
21 inequitable model capability. If the metric is validity-blind, the consequence is two-sided: tools
22 whose subgroup gap reflects benchmark composition rather than capability are penalized unfairly,
23 and tools whose lack of a stratified gap conceals true capability differences are cleared unjustifiably.
24 Both errors are borne disproportionately by the patients the audits are intended to protect, which
25 makes the question of whether stratified accuracy actually measures what it claims to measure a
26 deployment-grade concern, not just a measurement-theory one.

27 Suppose an FST-V/VI subgroup exhibits lower accuracy than an FST-I/II subgroup. Three explana-
28 tions are observationally indistinguishable from a stratified metric. *Capability difference*: models are
29 genuinely worse at the underlying construct on darker skin. *Difficulty composition*: the FST-V/VI
30 items happen to be harder along construct-relevant dimensions (rarer lesions, more advanced disease),
31 independent of FST per se. Critically, DDI is not paired across FST: the lesion-category distribution
32 differs across strata, so a per-stratum mean accuracy compares partially non-overlapping mixtures
33 of lesions, and a stratified gap admits a difficulty-composition explanation that cannot be ruled out
34 without item-level analysis. *Measurement non-invariance*: items in the two strata do not measure the
35 same construct in the same way, so the accuracy numbers are not on a common scale [Borsboom
36 et al., 2004]. Without separating these, any equity claim derived from a subgroup gap is unsupported.

37 *Do items in the DDI benchmark [Daneshjou et al., 2022] exhibit measurement non-invariance*
 38 *across FST groups when foundation vision-language models are evaluated on them, and if so, is that*
 39 *non-invariance attributable to clinical factors (lesion category, malignancy) rather than FST per se?*

40 We adapt amortized IRT calibration [Truong et al., 2025] to image-classification benchmarks, com-
 41 bine it with measurement-invariance analysis to produce per-item validity diagnostics, and apply
 42 the pipeline to DDI. We are deliberately narrow. DDI is sourced entirely from a single institution
 43 (Stanford), so our findings speak to measurement properties of this benchmark rather than to derma-
 44 tology AI evaluation in general, and any institution-level acquisition effects are absorbed into the FST
 45 contrast rather than partialled out. We do not causally identify FST as an independent driver of diffi-
 46 culty, since FST and lesion category are structurally entangled in DDI and further confounded with
 47 acquisition site. We report partial-identification analyses and adopt a hedged consequential-validity
 48 framing.

49 2 Related work

50 **IRT for ML benchmark evaluation.** Truong et al. [2025] introduced amortized IRT calibration for
 51 text benchmarks; we adopt their approach, port it to image embeddings, and extend it to measurement
 52 invariance, which they do not address. Polo et al. [2024] and Rodriguez et al. [2021] use IRT to
 53 construct compact benchmark subsets and rank NLP leaderboards, respectively; neither analyzes
 54 subgroup invariance.

55 **DIF and measurement invariance.** *Differential item functioning* (DIF) is the psychometric term
 56 for a test item whose probability of a correct response, conditional on the latent trait being measured,
 57 varies across groups of respondents—i.e., an item that does not function as the same measurement
 58 instrument across groups. Holland and Wainer [1993] provides the canonical DIF reference (Mantel-
 59 Haenszel, standardization), and Millsap [2011] grounds the configural/metric/scalar-invariance hier-
 60 archy we adopt. We invert the usual direction: rather than treating DIF as a property of human test
 61 items, we use it as a tool to audit ML benchmarks.

62 **Fairness and validity in medical AI.** Daneshjou et al. [2022] introduced DDI to enable FST-
 63 stratified evaluation and documented substantial accuracy gaps; subsequent work has expanded the
 64 catalogue [Groh et al., 2021]. Wang et al. [2025] argue more broadly that fairness metrics conflate
 65 disparate kinds of difference. We are not aware of prior work applying item-level measurement-
 66 invariance analysis to medical AI benchmarks; we close that gap.

67 3 Methods

68 **Course-material connections.** The pipeline below draws on three threads from Truong and Koyejo
 69 [2026]. The Rasch model and its amortized calibration variant are the item-response and amortized-
 70 factor models of Chapter 2 (Foundations of Measurement). Joint MLE for θ and ϕ follows the
 71 parameter-estimation treatment of Chapter 3 (Learning). The aggregate measurement-invariance test
 72 (§3) operationalizes Chapter 6’s framing of differential item functioning and construct-irrelevant
 73 variance as validity diagnostics, with the construct/criterion validity vocabulary used as the conse-
 74 quential frame. The LLTM decomposition (§3) is a structured-latent-variable extension that bridges
 75 Chapters 2 and 6, casting the residual-FST coefficient as a bridge between benchmark instrumentation
 76 and validity audit.

77 **Respondent selection.** We plan to query 12–20 foundation vision and vision-language models
 78 spanning architectural families. Contrastive: CLIP, SigLIP, BiomedCLIP. Self-supervised: DINOv3.
 79 Open multimodal: LLaVA-class, Qwen-VL, InternVL. Closed APIs: GPT-4o-class, Claude, Gemini.
 80 Classical IRT assumes exchangeable respondents; foundation models share pretraining corpora and
 81 architectures and are not random draws from any population. We therefore weaken the population
 82 interpretation, and ability estimates should be read as referring to the currently extant frontier
 83 multimodal ecosystem. The primary endpoint (below) is an aggregate item-level statistic robust to
 84 this weakening.

85 **Query protocol.** Per (model, item): single response, binary task framing (benign vs. malignant),
 86 single fixed prompt version-controlled in `config/protocol.yaml`, temperature 0 where supported,
 87 structured-output extraction with a GPT-4o-mini fallback parser. *Refusals are an analytic outcome*
 88 *category, not coerced into a default label.* A paraphrase-perturbed subset measures prompt sensitivity;
 89 $K = 3$ samples per (model, item) on a stratified subset estimate API non-determinism.

90 **Amortized Rasch calibration.** Let $X_{ij} \in \{0, 1\}$ indicate whether model j answered item i
 91 correctly. Under the Rasch model, $P(X_{ij} = 1 \mid \theta_j, b_i) = \sigma(\theta_j - b_i)$ [Truong and Koyejo, 2026,
 92 Ch. 2]. Following Truong et al. [2025], we amortize item difficulty as $b_i = f_\phi(e_i)$ where e_i is a
 93 fixed image embedding and f_ϕ is a small MLP. Primary embedding model: **BiomedCLIP** [Zhang
 94 et al., 2025] (dermatology-relevant pretraining). **DINOv3** [Siméoni et al., 2025] is used in parallel
 95 as a domain-agnostic robustness check. We jointly optimize $\{\theta_j\}$ and ϕ by maximizing the masked
 96 binary log-likelihood with Adam. Validation: (i) held-out item predictive AUC (20% items); (ii)
 97 Spearman correlation against traditional joint-MLE Rasch on items with sufficient response density;
 98 (iii) infit/outfit mean-squares with flagging threshold $[0.7, 1.3]$.

99 **Aggregate measurement invariance (primary endpoint).** Standard DIF treats persons as the
 100 grouping variable [Truong and Koyejo, 2026, Ch. 6]; in our setup FST is an item-level attribute,
 101 inverting that configuration. We therefore frame the primary analysis as a measurement-invariance
 102 test on item subsets. After fitting the amortized Rasch model, we residualize $\hat{b}_i$ against lesion category
 103 and malignancy by OLS and compute

$$\Delta = \overline{\hat{b}_i^{\text{resid}}}_{i \in \text{FST V-VI}} - \overline{\hat{b}_i^{\text{resid}}}_{i \in \text{FST I-II}}. \quad (1)$$

104 Uncertainty: non-parametric bootstrap over items, $B = 2000$ resamples. **Decision rule:**

$$|\Delta| \geq 0.5 \text{ logits} \quad \text{AND} \quad \text{bootstrap 95\% CI of } \Delta \text{ excludes zero} \quad \Rightarrow \quad \text{meaningful FST non-invariance.} \quad (2)$$

105 The 0.5-logit threshold reflects the conventional small-DIF cutoff in psychometric practice;
 106 we additionally report 0.3 and 0.7 as sensitivity checks but treat 0.5 as binding. *Configural-invariance*
 107 *check:* we refit the amortized model separately on FST-I/II and FST-V/VI subsets and report the
 108 Spearman correlation of model abilities; $\rho > 0.9$ is consistent with configural invariance (same
 109 construct ranks models the same way) even when difficulties shift.

110 **LLTM mechanism decomposition and metadata-only baselines.** We fit nested specifications
 111 of the form $b_i = \beta_0 + \beta^\top \mathbf{x}_i$ with increasing feature sets: **M1** lesion category; **M2** +malignancy;
 112 **M3** +FST group dummies (I/II reference). M1 and M2 double as *metadata-only baselines*: they
 113 predict per-item difficulty from clinical attributes that any reader of the dataset card could enumerate
 114 without running a single model query, and their R^2 against $\hat{b}_i$ is the share of difficulty variance that a
 115 chart-review-style audit would already explain. The headline mechanism estimate is β_{FST} in M3: the
 116 residual FST coefficient after lesion and malignancy controls. We report (i) R^2 at each nesting step
 117 so the marginal contribution of the amortized embedding signal above metadata is explicit (the gap
 118 between R_{M2}^2 and an embedding-only regression R_{Emb}^2 is the finer-grained-insight contribution of IRT
 119 calibration), (ii) bootstrap 95% CIs on all coefficients, and (iii) the lesion-category $\times$ FST contingency
 120 table that drives the entanglement (§4). We do not include image-acquisition covariates in the LLTM
 121 because the DDI release does not expose acquisition metadata (device, site, photographer, capture
 122 protocol); residual acquisition confounding is treated qualitatively in the limitations.

123 **Mechanism and refusal-rate DIF.** (i) Spearman correlations of per-item residualized difficulty
 124 with image-derived properties computable from the JPEG (resolution, mean luminance, color-channel
 125 statistics) and with embedding-space distance to the FST-I/II centroid; these are descriptive, not
 126 a quantitative decomposition. (ii) Refusal probability by FST via logistic regression with cluster-
 127 bootstrap CI clustered on model. A refusal pattern that loads on FST is itself evidence of differential
 128 item functioning, and we report it as such.

129 **Respondent-side analysis: family-stratified ability and contrast (exploratory).** The re-
 130 spondent non-exchangeability noted in §3 is itself analyzable. We annotate each model in
 131 `config/models.yaml` with an architectural family (contrastive image-text, self-supervised im-
 132 age, open multimodal-instruct, closed multimodal-instruct) and treat family as a grouping variable

on the respondent side. We report: (a) the within-family mean and standard deviation of $\hat{\theta}_j$ and a one-way ANOVA F -statistic for family on $\hat{\theta}$, partitioning between-family from within-family variance in ability; (b) the family-stratified primary contrast Δ_{family} , i.e., the FST difficulty shift recomputed with the respondent set restricted to each family in turn, so a finding that the V–VI vs. I–II gap is concentrated in a particular family (e.g., contrastive image-text models) is read off directly; (c) the Spearman correlation between per-item difficulty estimates obtained from disjoint respondent subsets defined by family, which tests whether different backbones produce a common item-difficulty ranking or partition into incompatible measurement instruments. This block is explicitly exploratory: with $J = 12\text{--}20$ models distributed across four families, within-family sample sizes are small and any family-level Δ carries wide uncertainty, so we report bootstrap CIs and decline a falsification threshold here.

4 Experimental setup

Data: DDI [Daneshjou et al., 2022] as the primary benchmark with FST I/II, III/IV, V/VI labels. We will report the lesion-category $\times$ FST and malignancy $\times$ FST contingency tables on the full DDI release up front, so the reader can read the entanglement structure (§3) off the data before seeing any IRT output. We do not use ISIC for inferential analysis (site-level photographic protocol differences would confound any FST signal); if used, ISIC supplies only pretraining signal for the amortized difficulty MLP in a two-stage design declared before the DDI fit. **Compute:** embedding extraction and open-weights inference run locally on a single GPU; closed-API inference scales as $J_{\text{api}} \times 656 \times P$ for P protocol variants. **Bootstraps:** $B = 2,000$ resamples for the primary CI; cluster-bootstrap on models for the refusal logit. **Metrics reported:** held-out Rasch AUC and infit/outfit; amortized-vs-traditional Spearman; primary Δ with bootstrap CI and configural-invariance Spearman; LLTM variance-explained at each nesting level (with M1/M2 read as metadata-only baselines) and β_{FST} CI in M3; mechanism Spearman; refusal-rate logit with cluster-bootstrap CI; family-stratified $\hat{\theta}$ and family-stratified Δ (§3).

5 Preliminary results (closed-API subpanel)

We report preliminary descriptive statistics from the four respondents that have completed the full DDI panel at the time of writing: two closed-API generative models—GPT-4o (gpt-4o-2024-11-20) and Claude Sonnet 4.5 (claude-sonnet-4-5)—and two contrastive zero-shot models—CLIP ViT-L/14 (openai/clip-vit-large-patch14) and BiomedCLIP (microsoft/BiomedCLIP-PubMedBERT_256-vit_base_patch16_224). A third contrastive respondent (SigLIP-L/16) and the open-weight VLM panel are still being collected; the pre-registered amortized Rasch primary endpoint (§3) is not yet estimated here. We present the conventional fairness-audit framing on the current four-respondent panel as the comparator against which the pre-registered IRT decomposition will later be evaluated, and three descriptive findings from this partial panel that already inform how that decomposition should be read.

Dataset entanglement (DDI metadata only). Before any model responses are introduced, the structural entanglement that motivates the LLTM decomposition (§3) is visible directly in DDI’s metadata. The 656 DDI items span 76 distinct lesion-category labels, and the marginal distribution differs sharply across FST strata. Selected examples: basal-cell carcinoma is concentrated in FST-I/II and FST-III/IV (7 and 34 items respectively, vs. *zero* in V–VI); dysplastic nevus is 2 / 14 / 0; mycosis fungoides is 3 / 3 / 26 (heavily concentrated in V–VI); eczema-spongiotic-dermatitis is 0 / 0 / 4 (V–VI only). A per-FST accuracy aggregate therefore compares partially non-overlapping mixtures of lesion categories, not a paired comparison on a common item pool. The full 76×3 contingency table is reported in `artifacts/contingency/lesion_x_fst.tex`.

The malignancy base rate also varies non-monotonically across FST groups (Table 1): 23.6% in FST-I/II, 30.7% in FST-III/IV, and 23.2% in FST-V/VI. This pattern is incompatible with the simplest "FST-V/VI has more advanced disease" interpretation of an accuracy gap, since by-stratum prevalence peaks in III–IV.

Baseline FST-stratified accuracy. Table 2 reports per-model accuracy under the binary benign-vs-malignant protocol, stratified by FST group. The two generative respondents (GPT-4o, Claude)

Table 1: Malignancy $\times$ FST contingency on the full DDI release. The non-monotonic per-stratum malignancy base rate (peaking in III–IV) rules out a simple prevalence-shift explanation for the V–VI accuracy gap.

Malignant	FST I–II	FST III–IV	FST V–VI	Total
False	159	167	159	485
True	49	74	48	171
Total	208	241	207	656

Table 2: Baseline FST-stratified accuracy (four-respondent preliminary panel: 2 closed-API generative, 2 contrastive zero-shot). Per-model proportion of items answered correctly under the binary benign-vs-malignant protocol. The right-most column is the V–VI minus I–II accuracy gap that conventional fairness audits would report; the pre-registered Rasch-difficulty analysis (§3) tests whether this gap reflects differential capability or differential item difficulty. Item counts per FST group in parentheses.

Model	N	FST I–II	FST III–IV	FST V–VI	Gap (V–VI – I–II)
GPT-4o	656	0.596 (208)	0.631 (241)	0.541 (207)	−0.055
Claude Sonnet 4.5	656	0.611 (208)	0.602 (241)	0.546 (207)	−0.065
CLIP ViT-L/14	656	0.721 (208)	0.647 (241)	0.691 (207)	−0.030
BiomedCLIP	656	0.562 (208)	0.598 (241)	0.517 (207)	−0.046

show V–VI minus I–II gaps of −5 to −7 percentage points, with a non-monotonic pattern across FST (III–IV intermediate or slightly higher than I–II rather than a linear decline). BiomedCLIP shows a comparable −5 pp V–VI gap. CLIP ViT-L/14 is an outlier: its 68.4% top-line accuracy exceeds the generative models’, and its V–VI gap is only −3 pp—a pattern we explain below as a degenerate-classifier artifact rather than capability.

Per-model prediction-distribution patterns (Figure 1). The accuracy table conceals dramatically different per-model calibration regimes. We plot the proportion of “benign” vs. “malignant” predictions each respondent emits per FST stratum, with the true malignancy rate overlaid as a black tick (Figure 1). Three patterns are visible:

- **GPT-4o and Claude Sonnet 4.5:** over-predict malignancy by 15–25 percentage points roughly uniformly across FST (40–47% predicted vs. 23–31% true). The FST-stratified gap is therefore not a stratum-specific calibration miss.
- **BiomedCLIP:** over-predicts malignancy more aggressively still (50–58% predicted vs. 23–31% true).
- **CLIP ViT-L/14:** systematically *under*-predicts malignancy (5.3%/17.0%/13.5% across FST-I/II, III/IV, V/VI vs. true 23.6%/30.7%/23.2%). In FST-I/II it is nearly a constant-“benign” classifier. Its 68% top-line accuracy is almost entirely an artifact of DDI’s 78% benign-prevalence baseline. Cohen’s κ vs. every other respondent on the parseable subset is near zero (§5), consistent with degenerate behavior masked by a label-prevalence-friendly metric.

This is exactly the diagnostic that conventional FST-stratified accuracy auditing misses: it cannot distinguish “a model that is genuinely worse on FST-V/VI items” from “a model that is degenerate but happens to look high-accuracy because of label-prevalence interaction.” The IRT calibration explicitly characterizes degenerate respondents through near-zero $\hat{\theta}$ and inflated infit/outfit mean-squares (§3).

Inter-model agreement. Pairwise Cohen’s κ on the parseable subset ($n = 656$ each pair): GPT-4o vs. Claude Sonnet 4.5 = 0.47 (moderate); GPT-4o vs. BiomedCLIP = 0.28; Claude vs. BiomedCLIP = 0.31; GPT-4o vs. CLIP = −0.01; Claude vs. CLIP = 0.02; BiomedCLIP vs. CLIP = −0.01. The two generative respondents share a non-trivial latent signal with each other and a weaker but positive signal with BiomedCLIP, satisfying the construct-stability precondition for IRT (§3). CLIP’s near-zero κ against *every other respondent* is a separate diagnostic for its degenerate prediction

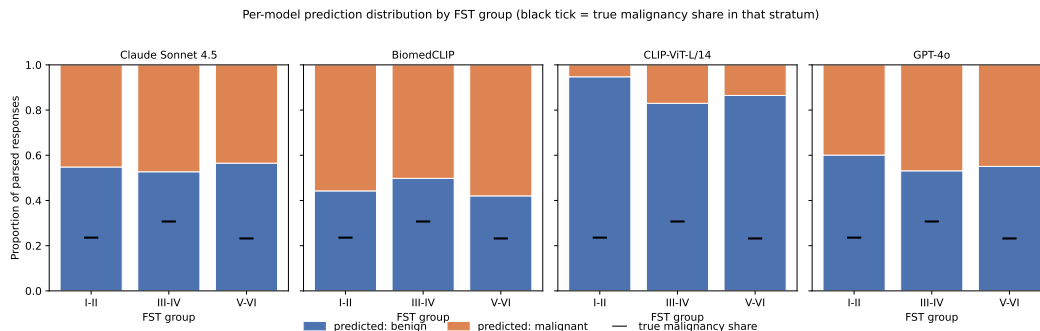


Figure 1: Per-model prediction-distribution by FST stratum. Stacked bars show the proportion of “benign” (blue) vs. “malignant” (orange) predictions on the parseable subset; the black horizontal tick on each bar marks the *true* malignancy share in that stratum. CLIP’s near-flat blue dominance in FST-I/II is a degenerate-classifier signature; the IRT calibration in §3 characterizes this through low $\hat{\theta}$ and inflated outfit.

pattern documented above. The configural-invariance check (§3) re-estimates the construct-stability claim once the full respondent panel is available.

Refusal behavior. All four current respondents return zero refusals and zero unparseable responses on the full panel (0/656 each); generative respondents follow the structured prompt template cleanly, and contrastive zero-shot respondents cannot refuse by construction. Refusal-rate DIF analysis (§3) therefore requires the open-weight generative VLM respondents (LLaVA, Qwen2-VL, InternVL2) to be informative; the current subpanel cannot test the pre-registered refusal-by-FST hypothesis.

What this does and does not show. These descriptive statistics are the inputs that conventional FST-stratified fairness audits would report and conclude on directly: “two frontier closed-API vision-language models show a 5–7 percentage-point accuracy drop on FST-V/VI relative to FST-I/II.” The pre-registered analysis (§3) is the test of whether that statement, once aggregated across the full respondent panel and adjusted for item-level difficulty composition, survives. We draw no inferential conclusion from the partial panel here; we report it as the baseline against which the pre-registered IRT contrast will be evaluated.

Pre-registration deviations. We record one deviation from config/models.yaml as frozen at prereg-v1. The originally pre-registered Gemini respondent, gemini-2.0-flash, was deprecated by Google before our query phase completed and now returns 404. We substituted gemini-3.1-flash-lite—the production-tier Flash-family model in the current Gemini line—as the closest analog. (A briefly attempted intermediate substitution to gemini-3.1-pro-preview was abandoned after that model’s ≈ 250 requests/day quota proved insufficient to complete the panel.) No other deviations.

6 Timeline and risks

This is a solo project; all weekly tasks below are owned by the single author. **Week 1 (setup):** finalize the respondent list with family annotations, fix the query protocol and parsing rules, build the DDI loader, BiomedCLIP and DINOv3 embedding pipelines, and the IRT codebase, and run amortized calibration on a 50–100 item pilot subset to verify the pipeline end-to-end. **Week 2 (query and fit):** query respondents on all DDI items under the primary protocol, run paraphrase-perturbed and $K = 3$ subsets, fit amortized and traditional Rasch, and report fit statistics. **Week 3 (analysis):** aggregate Δ with bootstrap, configural-invariance Spearman, LLTM nested fits, mechanism Spearman, and refusal-rate DIF. **Week 4 (writing):** draft results, generate figures, run sensitivity sweeps surfaced during writing, and submit.

Risks and mitigations. *Respondent non-exchangeability.* Foundation models are not random population draws. *Mitigation:* weakened interpretation in §3, and family-stratified Δ reported.

247 *Acquisition confounding cannot be partialled out.* DDI is largely Stanford-derived and acquisition
 248 conditions plausibly covary with patient flow and therefore with FST, but the release does not expose
 249 acquisition metadata. *Mitigation:* we treat this as an irreducible limitation rather than something
 250 the LLTM resolves; if M3 reports a meaningful β_{FST} , we explicitly flag that this residual could still
 251 reflect acquisition rather than FST. *Refusal-rate variation.* Closed APIs may refuse at FST-correlated
 252 rates. *Mitigation:* refusals are an analytic category and refusal-rate-by-FST is itself an invariance
 253 signal we report. *Amortization instability.* *Mitigation:* validation thresholds (§3); if not met, fall
 254 back to traditional joint-MLE Rasch with wider uncertainty and note the fallback in limitations. *API*
 255 *non-determinism.* *Mitigation:* $K = 3$ samples on a stratified subset estimate a within-model noise
 256 floor on Δ .

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